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Leveraging Oil and Gas Legacy for Advanced CCS in Japan: Realistic CO_2 Storage Simulation in the Higashi-Niigata Gas Field

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Abstract

Japan is advancing carbon dioxide capture and storage (CCS) initiatives to achieve carbon neutrality by 2050, identifying depleted oil and gas fields as promising storage sites. Among these, the Higashi-Niigata gas field – known for its long production history and extensive subsurface data — has been selected as a candidate site under the Advanced CCS Projects program led by the Japan Organization for Metals and Energy Security (JOGMEC).

Due to Japan's unique geological conditions, storage reservoirs are not always as favorable as those found in other countries. However, the Higashi-Niigata field, being a depleted gas field with a long production history and abundant data, offers a distinct advantage over data-scarce aquifers for precise storage planning. In this study, Japan Petroleum Exploration Co., Ltd. (JAPEX) and MITSUBISHI GAS CHEMICAL COMPANY, INC.(MGC) jointly conducted a comprehensive reservoir evaluation to assess the feasibility of CCS implementation at the Higashi-Niigata field. Before initiating a reservoir evaluation, the seal integrity and fault stability were evaluated using geological and production history data. A geological model was constructed, and a reservoir simulation model was developed through history matching, despite the complexity of the interbedded sand and mud formations characteristic of the field.

By leveraging extensive subsurface data and incorporating multiple geological realizations calibrated through history matching, the study achieved a highly accurate CO_2 storage plan. This plan enabled the quantitative assessment of injection performance, pressure evolution, and storage capacity. The inclusion of case studies involving pressure relief wells further highlighted optimization strategies for preserving reservoir integrity and enhancing storage potential.

The study demonstrated that the Higashi-Niigata gas field is a suitable and strategic candidate for CCS in Japan, with reliable injectivity, capacity, and integrity confirmed through simulation and data analysis.

Background and Objectives

In October 2020, the Government of Japan announced its target of achieving carbon neutrality by 2050, aiming for net-zero greenhouse gas (GHG) emissions. Furthermore, in April 2021, an interim goal was set

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to reduce GHG emissions by 46% compared to 2013 levels by 2030. These targets have accelerated efforts toward achieving a decarbonized society.

As part of this initiative, the Act on Carbon Dioxide Storage Business was enacted by the Japanese Parliament in May 2024, establishing a regulatory framework for carbon dioxide capture and storage (CCS) operations. Additionally, the Japan Organization for Metals and Energy Security (JOGMEC) launched support for operators under its 'Advanced CCS Projects' program from 2023, aiming for commercialization by 2030 (see Figure 1).

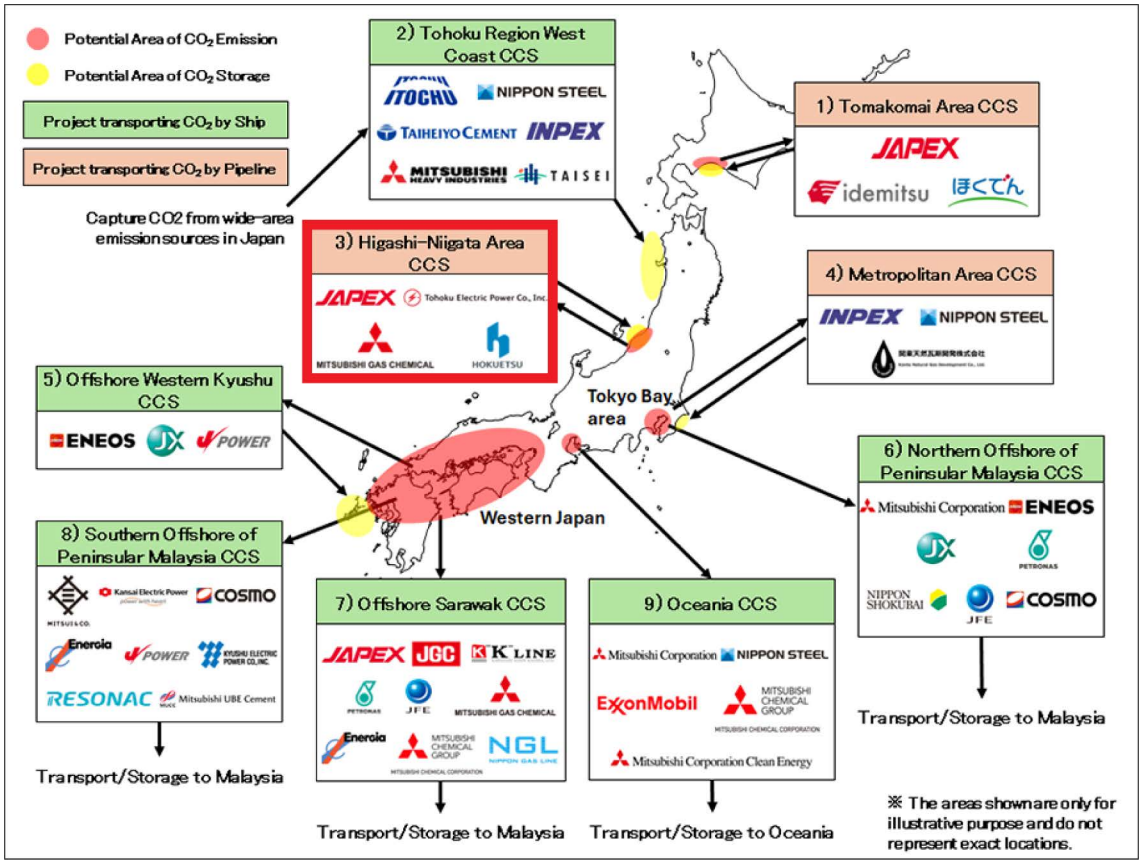


Figure 1—Japanese Advanced CCS Projects and the red box denotes this project[1]

In this context, Japan Petroleum Exploration Co., Ltd. (JAPEX) is planning CCS and carbon dioxide capture, utilization, and storage (CCUS) projects to contribute to Japan's overall CO_2 reduction target of approximately 760 million tons. The Higashi-Niigata region, located along the Japan Sea, is one of Japan's major oil and gas production hubs and is surrounded by a concentration of CO_2 emission sources such as thermal power plants and chemical factories, with total annual emissions estimated at several million tons. Leveraging these geographical and industrial characteristics, a 'hub-and-cluster' CCS project is being conceptualized in this region, in which CO_2 collected from multiple emitters is aggregated via pipelines and injected underground for storage.

This paper reports on the results of a reservoir study promoted by a five-company consortium, including JAPEX, as part of the Higashi-Niigata CCS project selected by JOGMEC in 2024 as one of the Advanced CCS Projects.

In evaluating the feasibility of the Higashi-Niigata CCS project, Japan's unique geological conditions present significant challenges. CCS target reservoirs in Japan are not necessarily large-scale or of exceptionally high quality, as seen in some other countries, and thus may not be inherently favorable for CCS deployment. However, as shown in Table 2, implementing CCS in depleted oil and gas fields offers

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numerous advantages comparable to those of saline aquifers, most notably the availability of abundant subsurface data. The Higashi-Niigata depleted oil and gas field targeted in this study has greatly benefited from its extensive development and production history, providing a robust foundation for constructing a highly detailed and reliable storage plan. Leveraging this unique advantage, a detailed reservoir study of the Higashi-Niigata field was conducted to demonstrate the feasibility of the CCS project.

Table 2—Comparison of CCS Target Reservoir

	Depleted Oil/Gas Reservoirs	Saline Aquifers
Advantage	• Rich geological and fluid data • Proven long-term containment • Existing infrastructure available • Enhanced economic viability via EOR	• Large storage capacity • Wider site selection flexibility • Suitable for long-term, large-scale projects • Potential for advanced monitoring technologies
Challenge Examples	• Limited storage capacity • Risk of leakage from aging wells and infrastructure • Potential for rapid pressure buildup • Less suitable for major scale-up	• Limited geological and pressure data • Higher risks in pressure management and leakage • High costs for new infrastructure • No direct revenue generation
Project Examples	• Cranfield Project (USA)	• Sleipner Project (Norway) • Gorgon Project (Australia)

Technical workflow

Overview of the Target

Japan Petroleum Exploration Co., Ltd. (JAPEX) and MITSUBISHI GAS CHEMICAL COMPANY, INC. (MGC) discovered the Higashi-Niigata gas field, located in Niigata Prefecture along the Japan Sea coast, in 1959. Over 40 wells have been drilled in this field. After approximately 50 years of development and production, the field is now in a late-mature stage.

In recent years, gas and oil production have declined due to pressure depletion and water production, prompting growing interest in the potential for CO_2 injection into depleted reservoirs.

The Higashi-Niigata field comprises multiple reservoirs, each overlain by independent seal formations. One of these multiple reservoirs has the highest cumulative production volumes, and this reservoir is considered a promising candidate for CO_2 storage.

CO_2 Storage Suitability in the Field

This section summarizes the CO_2 storage suitability of the Higashi-Niigata gas field from the perspectives of the reservoir, seal, and fault.

Reservoir. Among the multiple candidate formations, one was selected as the primary storage target, while the others were evaluated as secondary candidates. These formations are in a supercritical condition and exhibit favorable horizontal continuity and pressure communication, suggesting their suitability for CO_2 storage. The reservoir simulations described later were conducted with the primary objective of confirming these characteristics.

Seal. Immediately above the target reservoirs lies a mudstone seal formation with a thickness of 50–120 meters, which has historically retained hydrocarbons, indicating strong sealing capacity for CO_2. Additionally, multiple overlying production layers may act as secondary or tertiary seals, thereby minimizing the risk of surface leakage.

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Faults. Seismic data and existing well information indicate the presence of multiple faults within the field. However, based on production data and observed fluid composition changes, the faults separating different reservoirs are interpreted to have low transmissibility. Therefore, the risk of CO_2 migration through faults is considered low if injection is conducted into the targeted reservoir formation.

Based on these findings, the reservoir exhibiting the most favorable properties within the Higashi-Niigata gas field was selected as the primary storage target, and reservoir simulations were subsequently conducted.

Reservoir Simulation Study

Model Overview. This section outlines the constructed reservoir model. The key model parameters are summarized in Table 3, and a schematic diagram of the model is in Figure 4.

Table 3—List of model parameters

Size	9 km × 11 km × 1800 m
Grid Dimensions	120 × 111 × 103
Number of Active Grid Cells	276,410

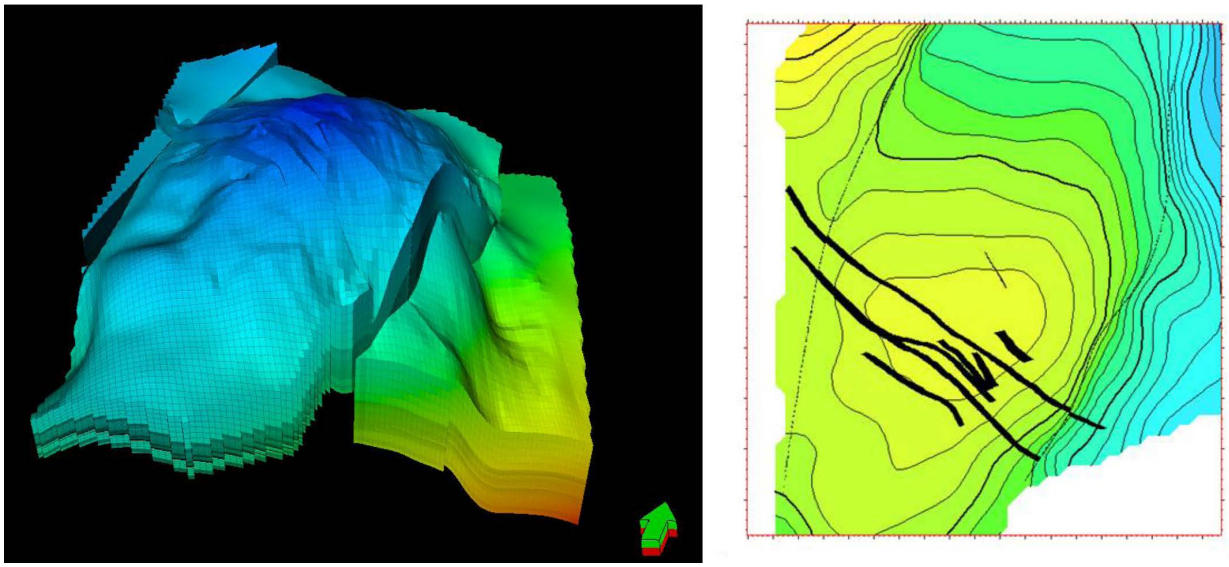


Figure 4—Overview of the reservoir model. (Left) 3D perspective view (Right) 2D depth map displaying the top reservoir surface.

To address geological uncertainty, 100 realizations were generated by varying key reservoir properties as follows:

- Volume of shale (Vsh): Spatial distribution estimated using geostatistical methods based on existing well data.
- Porosity: Derived using correlations with Vsh
- Permeability: Set using core-based porosity–permeability relationships and Flow Zone Indicator.

Subsequently, history matching was performed across all realizations. Based on the results, 10 realizations exhibiting good agreement with historical production data were selected for subsequent forecasting analyses.

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History Matching. To validate the model, history matching was conducted using commercial software with existing production data up to April 1, 2022. (Figure 5)

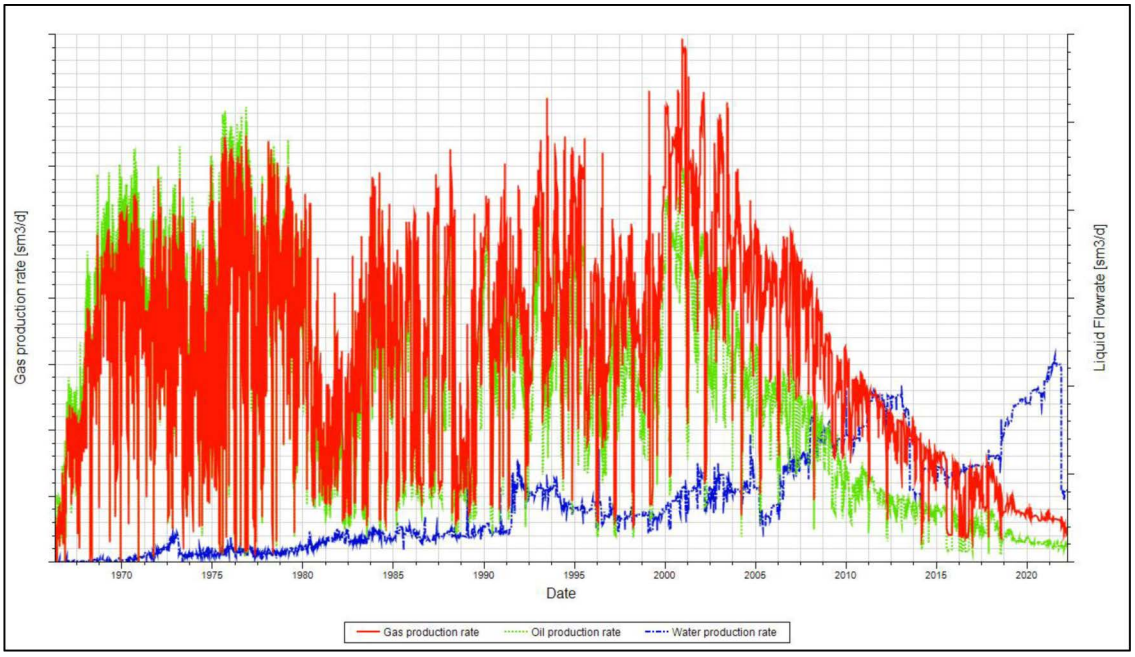


Figure 5—Production history in the Higashi-Niigata area

Figures 6 and 7 showed good agreement between the model and actual production data for gas and water. The Higashi-Niigata field consists of alternating sand and shale layers, forming a complex reservoir where reproducing water breakthrough behavior at each well is challenging. To evaluate the CO_2 storage capacity, a material balance matching approach was conducted to ensure consistency with observed performance. Ten realizations showing good agreement for pressure and water production were selected for use in the CO_2 injection case study.

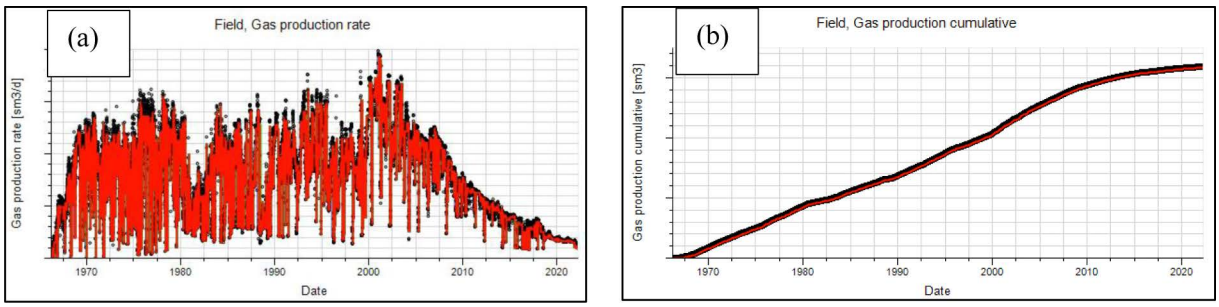


Figure 6—History matching results with all realizations (a) Gas rate (b) Cumulative gas production

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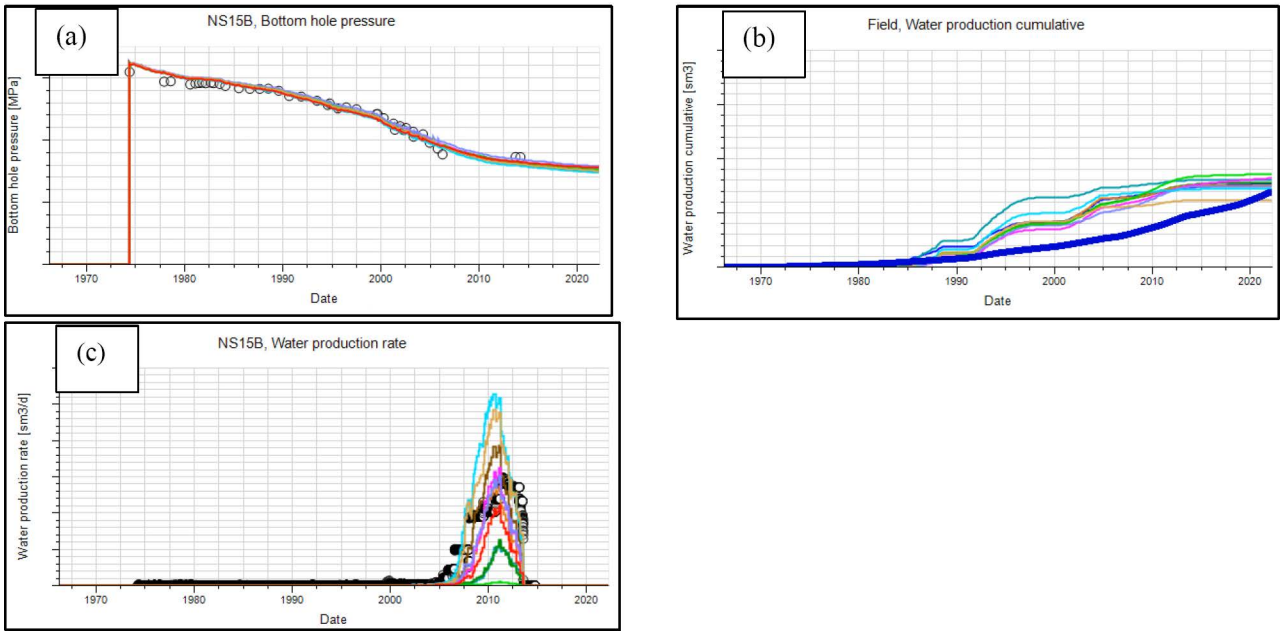


Figure 7—History matching with 10 realizations (a) Existing well bottom hole pressure (b) Cumulative water production from the field (c) Water production rate by the existing well

CO_2 Injection. Simulations were carried out to assess the behavior of CO_2 injection with the selected 10 realizations. The simulation conditions were as follows. (Table 8 and Figure 9)

Table 8—Simulation condition parameters

Number of Wells	**Case A**	**Case B**
	Injector : 3 **Relief well : 0**	**Injector : 3** **Relief well : 3**
Simulation Period	Jan 1, 2029 – Jan 1, 2130	
Assumed CO_2 Injection Period	Jan 1, 2029 – Jan 1, 2050	
Injection Pressure Limit	Initial Reservoir Pressure	

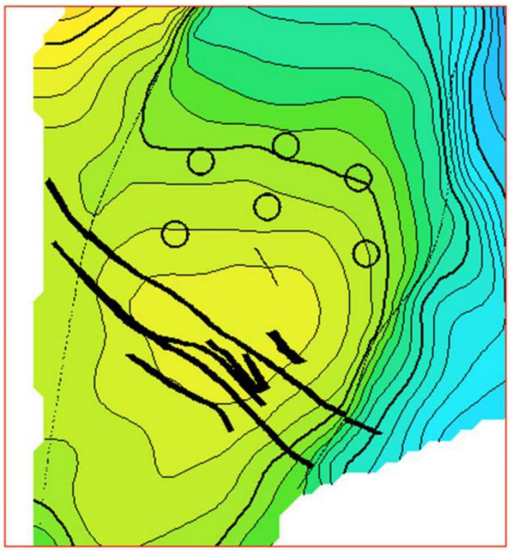


Figure 9—Injector location (circles) in 2D depth map

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Case A: CO2 Injection with 3 injectors

In the initial case, the CO_2 injection volume was evaluated by operating three injection wells. The placement of the injection wells is shown in Figure 10. It was designed to balance two objectives: (1) promoting residual trapping and dissolution by positioning wells deeper, and (2) minimizing deliverability uncertainty by situating wells closer to the central area, where extensive geological and production data are available. The results are presented in Figure 11 and 12.

- Around 2040, the reservoir pressure reached its upper limit, making it difficult to maintain the injection rate.
- Due to the density difference between CO_2 and the existing hydrocarbons, CO_2 tended to migrate toward the structural crest.

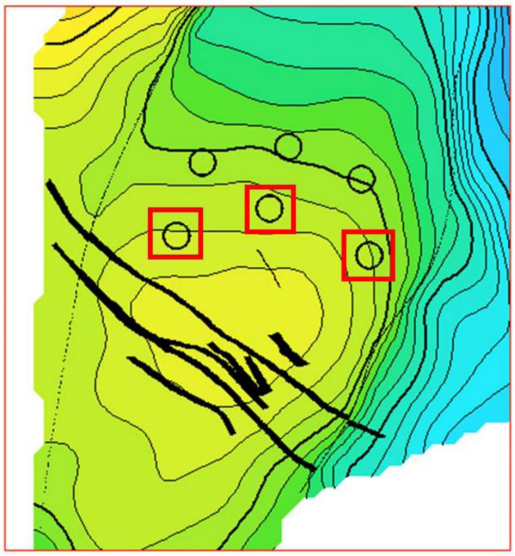


Figure 10—Injector (red boxes) in 2D depth map

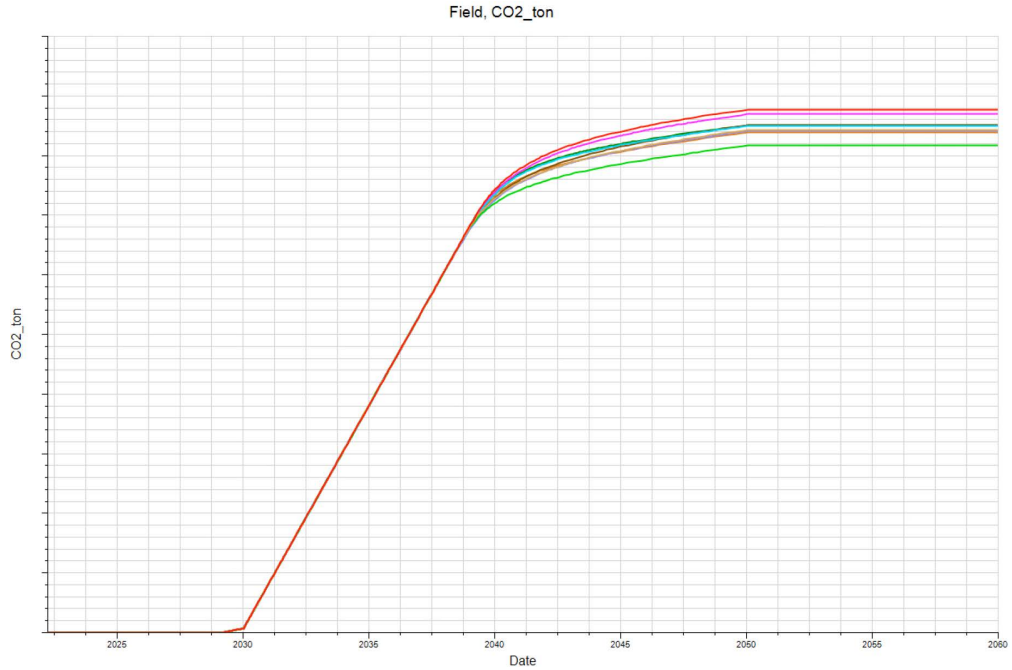


Figure 11—Cumulative CO2 injection volume with Case A

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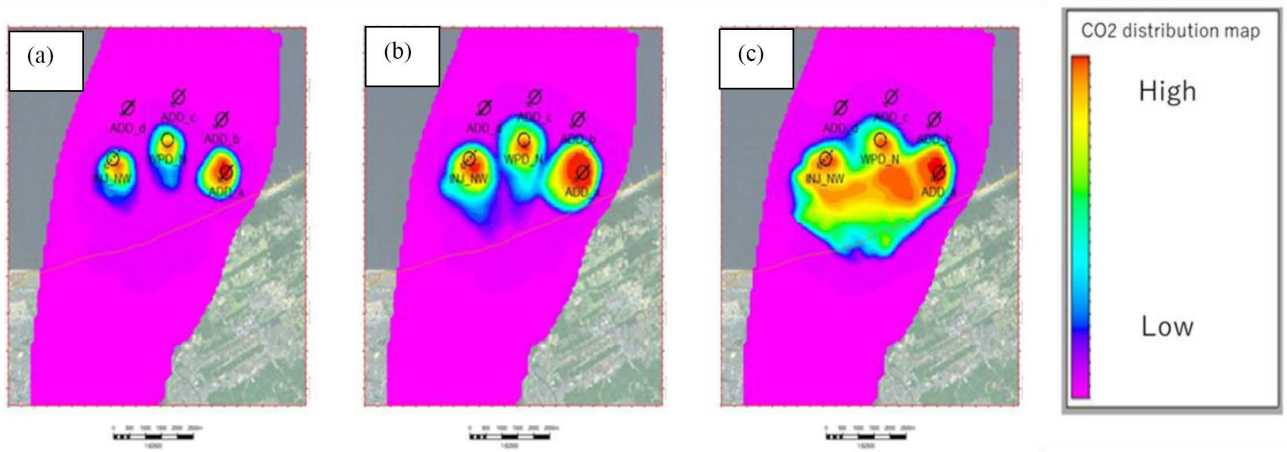


Figure 12—CO2 distribution map in the target reservoir with Case A
(a)5 years after injection (b) 10 years after injection (c) Simulation End

Case B: CO_2 Injection with Pressure Relief Wells

Since the evaluation was conducted using a reservoir model, a case incorporating pressure relief wells was examined as an option to secure greater storage capacity by incorporating water production into the material balance framework.

The locations of the injection wells and pressure relief wells are shown in Figure 16. The results are presented in Figure 17 and Figure 18. The placement of the pressure relief wells was also designed to balance two objectives: (1) distancing them from the injection wells to mitigate CO_2 breakthrough, and (2) situating them closer to the central area, where extensive geological and production data reduce deliverability uncertainty.

- It was observed that the CO_2 injection volume increased by approximately 25% to 30% compared to Case A.
- The time at which the reservoir pressure reached the upper threshold was delayed compared to Case A, confirming the effectiveness of the pressure relief wells (Figure 19).

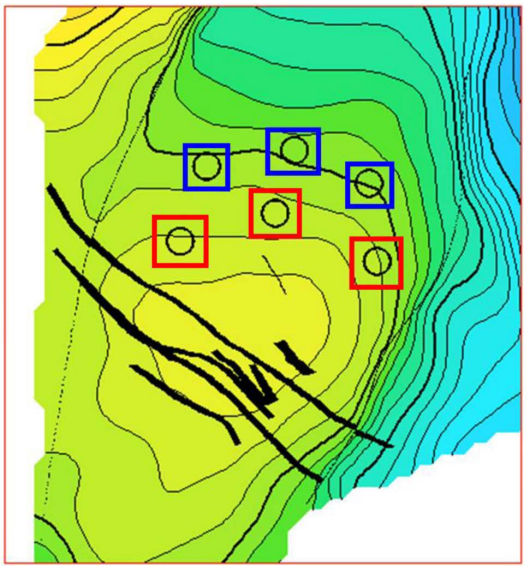


Figure 16—Additional wells in the 2D depth map Red box denotes injection wells, and blue one is pressure relief wells

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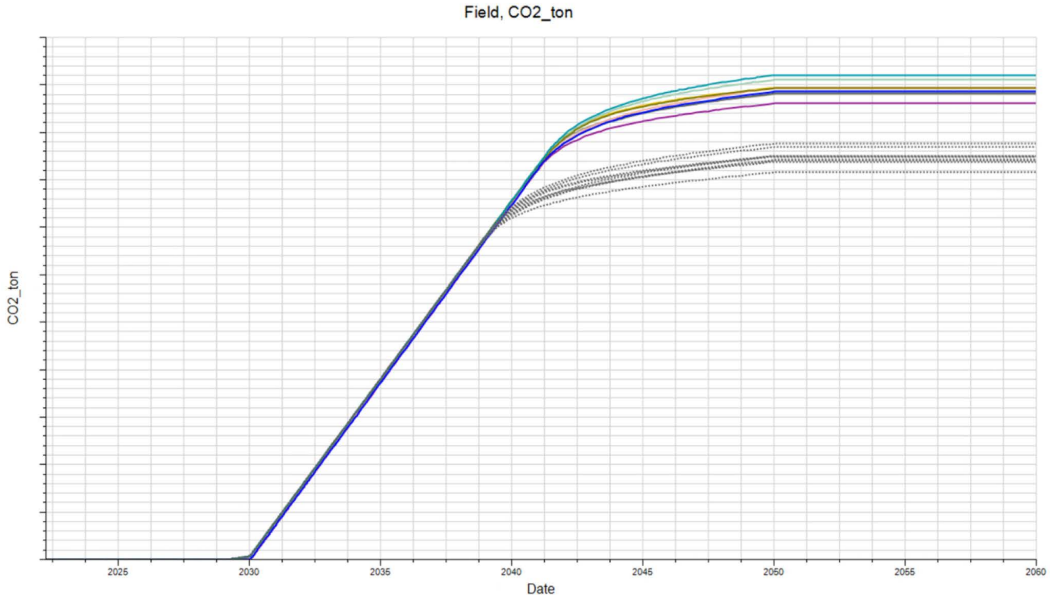


Figure 17—Cumulative CO_2 injection volume, with colored lines indicating Case B and gray dotted lines indicating Case A.

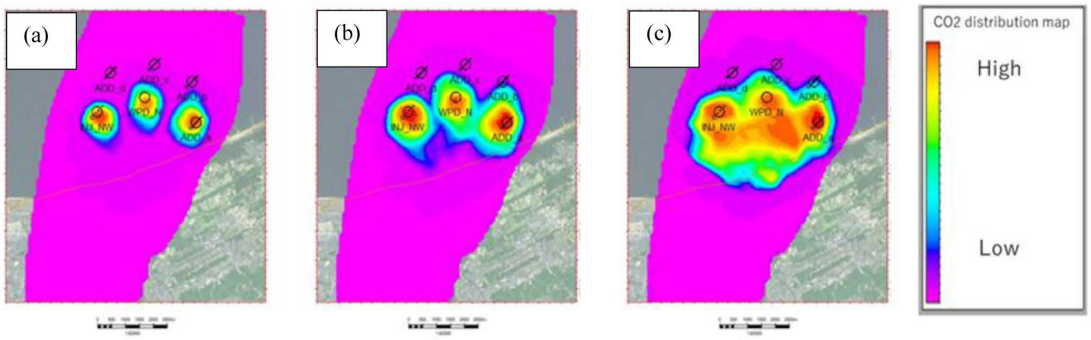


Figure 18—CO2 distribution map in the target reservoir with Case B
(a)5 years after injection (b) 10 years after injection (c) Simulation End

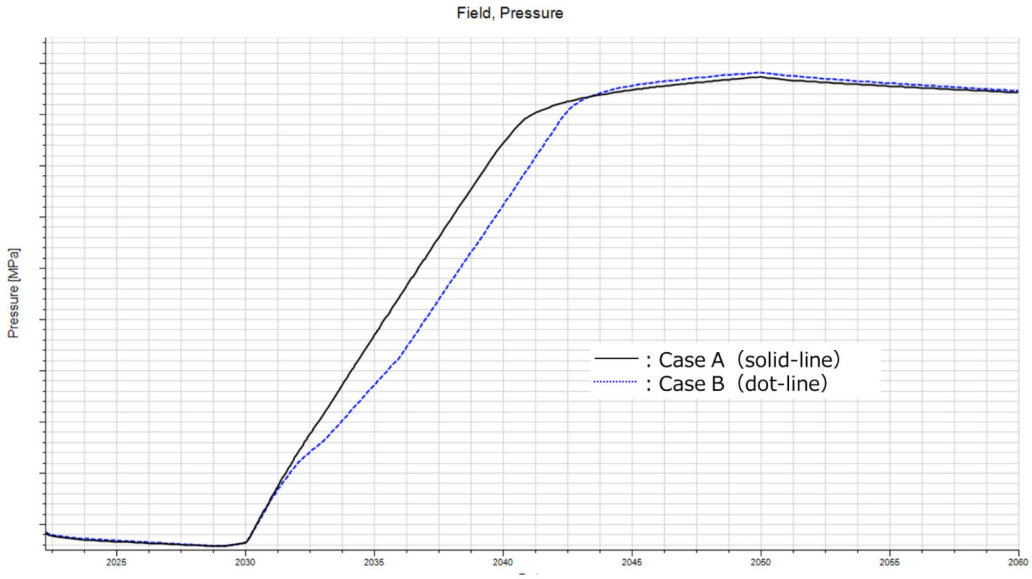


Figure 19. Average reservoir pressure trends

Figure 19—Average reservoir pressure trends

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Conclusion

This study assessed the CCS technical feasibility in the Higashi-Niigata gas field, a depleted gas reservoir in Japan. A comprehensive geological evaluation, reservoir modeling, and simulation study were conducted to characterize the CO2 storage potential and the injection behavior under various operational scenarios.

Leveraging the extensive subsurface data available from this mature gas field, the reservoir simulation study enabled the construction of a highly accurate CO_2 storage plan. By incorporating multiple geological realizations and conducting history matching, the study quantitatively assessed injection performance, pressure behavior, and storage capacity under various operational conditions. This methodology helped reduce the uncertainties typically associated with CCS projects in undeveloped formations and demonstrated the viability of site-specific, detailed storage planning. Case studies, including the scenarios utilizing pressure relief wells, offered valuable insights for optimizing injection operations while preserving reservoir integrity.

Given its long production history, extensive subsurface data, and comprehensive reservoir studies, the Higashi-Niigata field demonstrates not only high technical feasibility for CCS implementation but also significant potential as a strategic CCS hub within the "hub-and-cluster" framework promoted in Japan. These findings are expected to contribute to the establishment of large-scale CCS infrastructure and the achievement of national carbon neutrality targets.

Acknowledgements

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